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LATIHAN PERTEMUAN 3

PERCABANGAN, PERULANGAN, DAN OPERASI GIT LANJUTAN PADA DART

MOBILE PROGRAMMING

Link Github: <https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3>

Studi Kasus: Aplikasi Kalkulator BMI

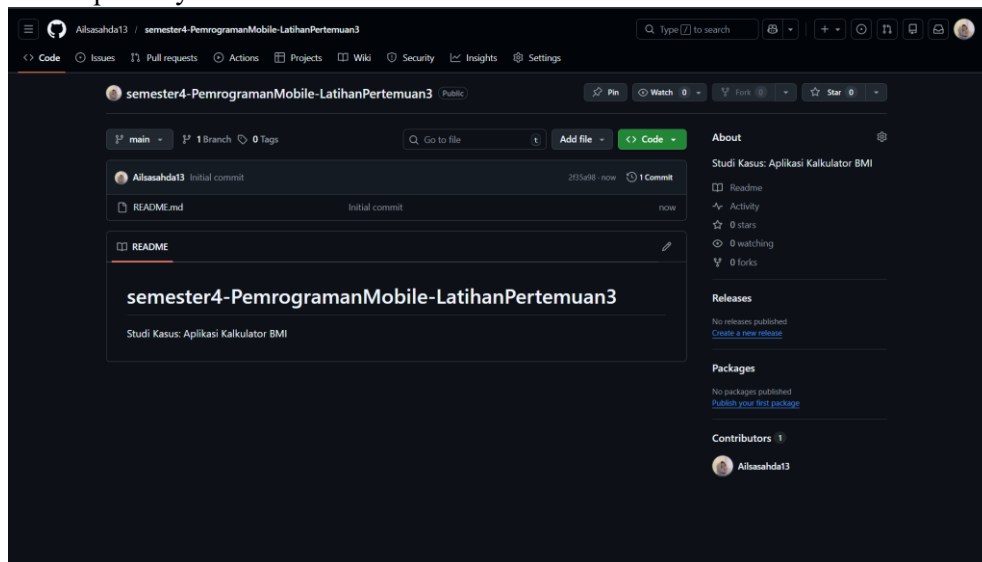
Kita akan mengembangkan aplikasi kalkulator BMI sederhana menggunakan Dart dengan menerapkan percabangan dan perulangan

Fitur Aplikasi:

- Input tinggi dan berat badan
- Kalkulator BMI
- Kategorisasi hasil (kurus, normal, gemuk)
- Riwayat Perhitungan (menggunakan perulangan)

Alur Kerja Git:

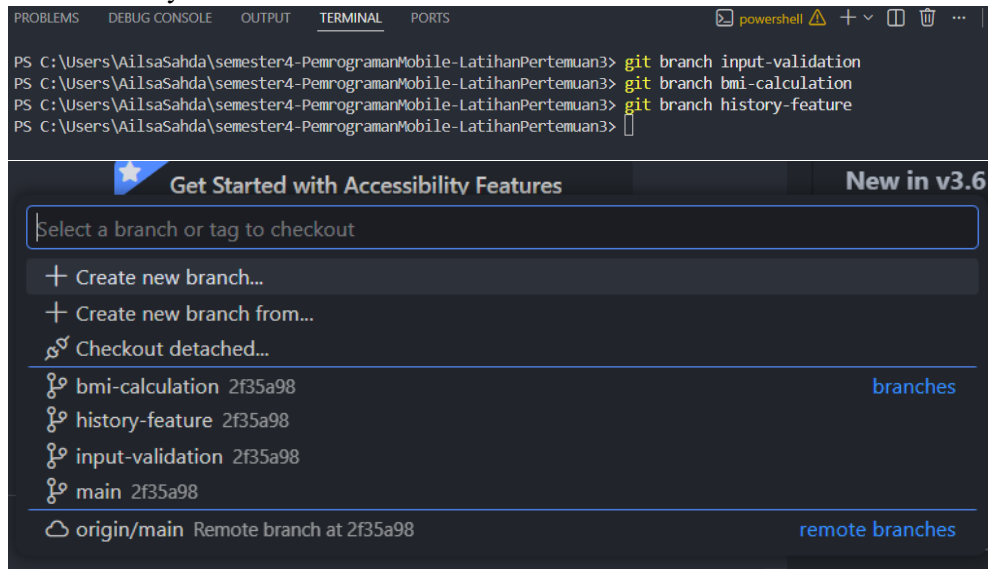
1. Buat repository baru di GitHub



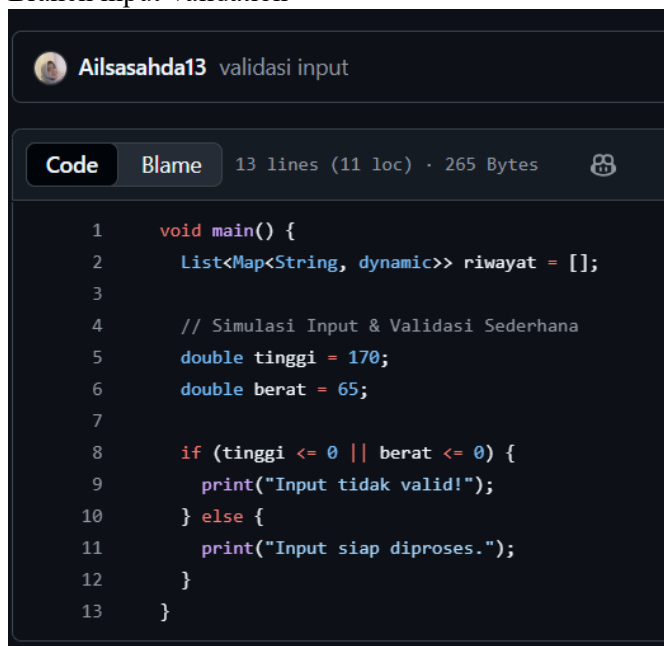
2. Clone repository ke local

```
PS C:\Users\AilsaSahda> git clone https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3.git
Cloning into 'semester4-PemrogramanMobile-LatihanPertemuan3'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\AilsaSahda>
```

3. Buat branch untuk setiap fitur:
- Branch input-validation
 - Branch bmi-calculation
 - Branch history-feature



4. Merge branch ketika fitur selesai
- Branch input-validation



- Branch bmi-calculation

Ailsasahda13 rumus perhitungan BMI

Code

Blame

27 lines (23 loc) · 600 Bytes



```
1 void main() {
2     double tinggiCm = 170.0;
3     double beratKg = 65.0;
4
5     if (tinggiCm > 0 && beratKg > 0) {
6         // Memanggil fungsi hitung (tanpa riwayat dulu)
7         hitungBMI(tinggiCm, beratKg);
8     }
9 }
10
11 void hitungBMI(double tinggiCm, double beratKg) {
12     double tinggiM = tinggiCm / 100;
13     double bmi = beratKg / (tinggiM * tinggiM);
14
15     String kategori;
16     if (bmi < 18.5) {
17         kategori = "Kurus";
18     } else if (bmi < 25) {
19         kategori = "Normal";
20     } else if (bmi < 30) {
21         kategori = "Gemuk";
22     } else {
23         kategori = "Obesitas";
24     }
25
26     print("BMI: ${bmi.toStringAsFixed(2)} - Kategori: $kategori");
27 }
```

```
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git commit -m "rumus perhitungan BMI"
[bmi-calculation 0a9dc6a] rumus perhitungan BMI
1 file changed, 27 insertions(+)
create mode 100644 bmiCalculation/bmicalculationjs3/bin/main.dart
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git push origin bmi-calculation
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 715 bytes | 178.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'bmi-calculation' on GitHub by visiting:
remote:   https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3/pull/new/bmi-calculation
remote:
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3.git
 * [new branch]      bmi-calculation -> bmi-calculation
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3>
```

- Branch history-feature

```

nothing to commit, working tree clean
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git checkout history-feature
Switched to branch 'history-feature'
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git branch
  bmi-calculation
* history-feature
  input-validation
  main
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git add .
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git commit -m "menambahkan riwayat data dan perhitungan"
[history-feature 3dde197] menambahkan riwayat data dan perhitungan
1 file changed, 38 insertions(+)
create mode 100644 bmiCalculation/bmicalculationjs3/bin/main.dart
PS C:\Users\AilsaSahda\semester4-PemrogramanMobile-LatihanPertemuan3> git push origin history-feature
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 882 bytes | 294.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'history-feature' on GitHub by visiting:
remote:   https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3/pull/new/history-feature
remote:
To https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3.git
 * [new branch]      history-feature -> history-feature

```


Ailsasahda13 menambahkan riwayat data dan perhitungan

Code

Blame

38 lines (33 loc) · 957 Bytes

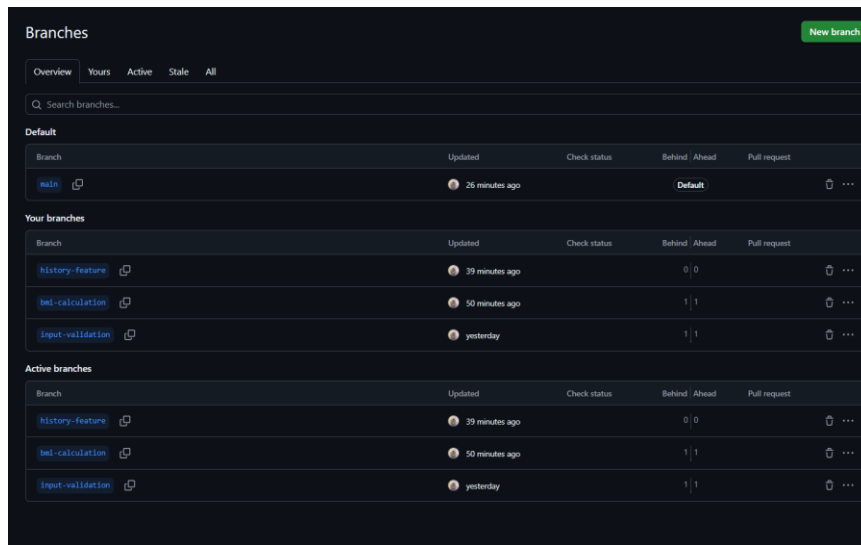


```

1  void main() {
2      List<Map<String, dynamic>> riwayat = [];
3
4      // Simulasi input sesuai jobsheet
5      hitungBMI(170, 65, riwayat);
6      hitungBMI(165, 75, riwayat);
7      hitungBMI(180, 85, riwayat);
8
9      // Perulangan untuk menampilkan riwayat
10     print("=== RIWAYAT HITUNG BMI ===");
11     for (var data in riwayat) {
12         print("Tinggi: ${data['tinggi']} | Berat: ${data['berat']} | BMI: ${data['bmi'].toStringAsFixed(2)} | Kategori: ${data['kategori']}");
13     }
14 }
15
16 void hitungBMI(double tinggiCm, double beratKg, List<Map<String, dynamic>> riwayat) {
17     double tinggiM = tinggiCm / 100;
18     double bmi = beratKg / (tinggiM * tinggiM);
19
20     String kategori;
21     if (bmi < 18.5) {
22         kategori = "Kurus";
23     } else if (bmi < 25) {
24         kategori = "Normal";
25     } else if (bmi < 30) {
26         kategori = "Gemuk";
27     } else {
28         kategori = "Obesitas";
29     }
30
31     // Simpan hasil ke riwayat
32     riwayat.add({
33         'tinggi': tinggiCm,
34         'berat': beratKg,
35         'bmi': bmi,
36         'kategori': kategori
37     });
38 }

```

- Branch



5. Push ke repository remote

Link github: <https://github.com/Ailsasahda13/semester4-PemrogramanMobile-LatihanPertemuan3>